Experiment-15

Error detection and correction using hamming code

Aim:

write a Program to implement error detection and correction using hamming code.

Creates senders Program:

1) Input to senders file should be a text of any length. Program should convert text to binary.

2) Apply hamming code concept in binary data.

3) Recievers Program should read the input from channel file.

code:

```
data = list(map(int, input("Enter the bits ")))
p1 = data[0] ^ data[1] ^ data[3]
p2 = data[0] ^ data[2] ^ data[3]
p3 = data[1] ^ data[2] ^ data[3]
code = [p1, p2, data[0], p3, data[1], data[2], data[3]]
if error:
    print("Error at bit:", error)
    code[error-1] ^= 1
    print("Corrected:", code)
else:
    print("no error detected")
```

3) If there is an error, display the position of the error.

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Result:

Thus the Program for error detection and correction was implemented successfully.